# CS 5594: BLOCKCHAIN TECHNOLOGIES

Spring 2023

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## BLOCKCHAIN OVERVIEW

# Outline

- Traditional Transactions
- Traditional Trust Models and Issues
- Why Blockchain? What is Blockchain?
- Blockchain Components and Useful Terminologies

# Traditional Transaction Model

$10,000

send(Alice, $10K)

notify TX received

**TRUSTED THIRD PARTIES**

| Account # | Name | Balance |
|---|---|---|
| 2136544 | Bob | $100,000 |
| 32136545 | Doge | $500,000 |
| 32136546 | Eve | $1,000,000 |

A

$10,000

B

| Account # | Name | Balance |
|---|---|---|
| 32136521 | Alice | $100,000 |
| 32136522 | Carol | $500,000 |
| 32136523 | Dave | $1,000,000 |

Ledger A

| Account # | Destination | Amount |
|---|---|---|
| 32136544 | 32136521 | - $60,000 |
| 32136544 | 32136521 | + $200,000 |
| 32136544 | 32136521 | - $10,000 |

Transaction records

Ledger B

| Account # | Destination | Amount |
|---|---|---|
| 32136521 | 32136544 | + $60,000 |
| 32136521 | 32136544 | - $200,000 |
| 32136521 | 32136544 | + $10,000 |

# Traditional Transaction Issues

Pros
- **Trust** & Integrity
- Centralized
- Standardized processes & procedures

Cons
- Security, Privacy, **Trust**
- Intermediation
  - Expensive, efficiency
- Centralized
  - Counterparty risk/fraud
  - Single point of failure

## TRUSTED THIRD PARTIES

| Account # | Name | Balance |
|---|---|---|
| 2136544 | Bob | $100,000 |
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A

Send $10,000

$10,000

B

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# Traditional Transaction Issues

- Centralized systems suffer from many factors
  - **Trust**
  - Security and Privacy
  - Fraud, single point of failure
  - Operational cost to maintain trusted entities

# What is Trust?

- All decisions we make everyday are based on a degree of trust
  - Driving car
  - Buying groceries
  - Online purchase
- Trust reduces operational cost
  - Without trust, one must verify the reliability of everything
  - Impossible to verify everything

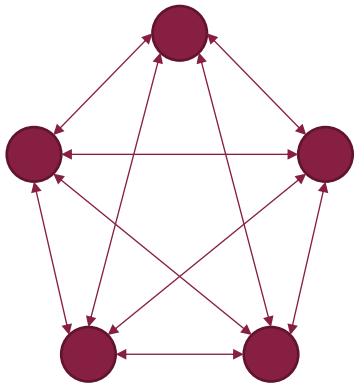
# What is Trust?

- Trust is gray-scale. There are different degrees of trust
  - Full trust
  - Semi-trust
  - Weak trust
- Trust is a two-sided coin
  - A complex psychological state combined of rational and emotional factors
  - Acceptance of uncontrolled and unquantified risk
  - “Trust begins where prediction ends.” – David Lewis and Andrew Weigert

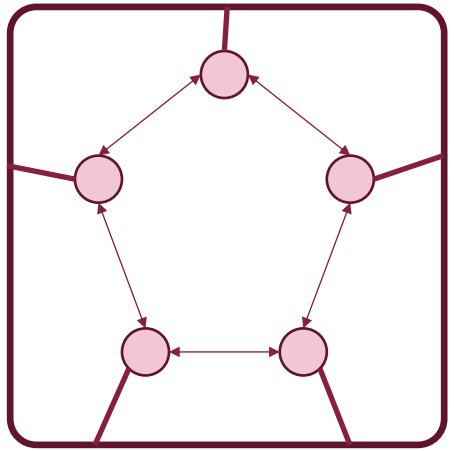
# Conventional Trust Models

Three trust architectural models:

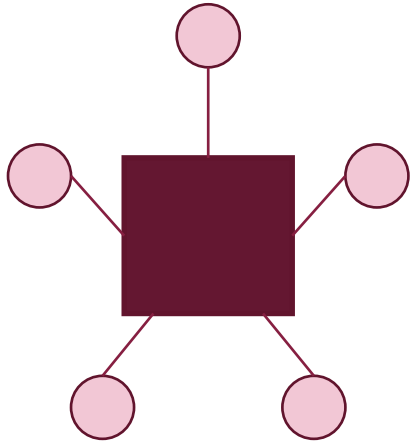
- **<u>P2P</u>**: I trust you because of *you*
- **<u>Leviathan</u>**: I trust you because of legal contracts established by trusted authorities
- **<u>Intermediary</u>**: I trust you because of trusted platform we both operate on



P2P



Leviathan

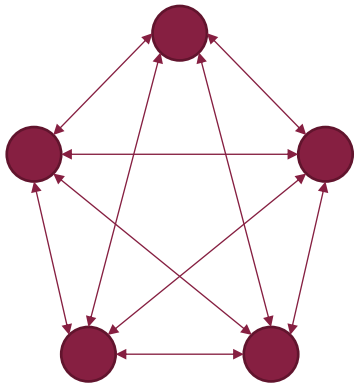


Intermediary

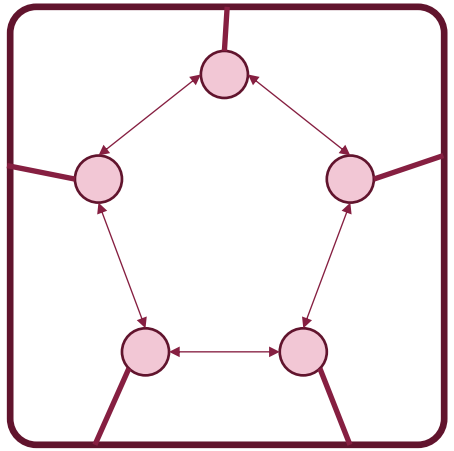
# Conventional Trust Models

**Three trust components per transaction**

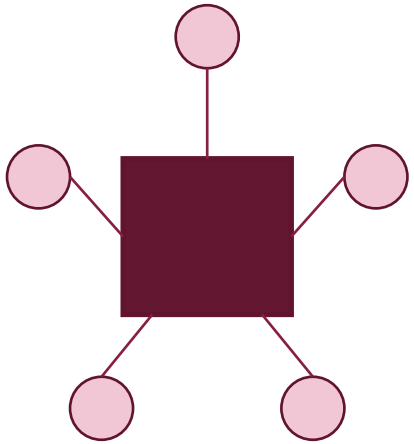
- Counterparty (P2P)
- Dispute resolution mechanism (CA)
- Intermediary



P2P



Leviathan



Intermediary

# A Crisis of Trust

- Trust forms the ways we interact and behave
  - High-trust societies are powerful and outperform low-trust societies
- **However, trust is in crisis!!!**
  - Trust level is decreasing over recent years
  - 80% of Americans do not trust government, 67% of them do not trust each other

# A Crisis of Trust

*"The root problem with conventional currency is all the trust that's required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust."*

- Satoshi Nakamoto

# A Crisis of Trust

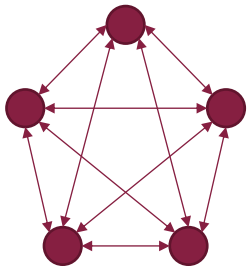
Trust fails because of

- **Direct violation:** non-reputed organization
  - Lend somebody money
- **Opportunistic behavior**: When benefits outweigh trust factors
  - Facebook–Cambridge Analytica scandal
- **Systemic collapse**: Unwanted behaviors (compromised, corrupted)
  - Equifax data breach, Apple iCloud, Sony PlayStation Network

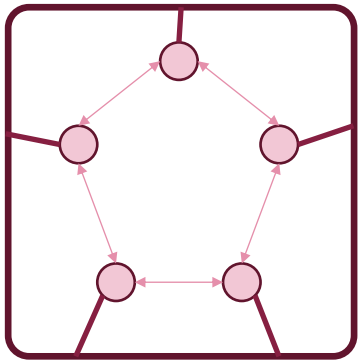
<u>Extremely hard to restore/recover trust when it fails</u>

# Blockchain: A Revolution of Trust

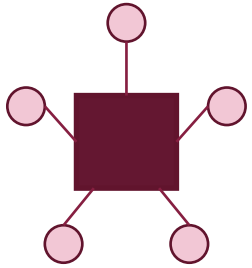
- **Blockchain**: a revolution of trust
  - "Trustless trust" – Reid Hoffman (LinkedIn founder)
  - A trusted network without trusting anyone



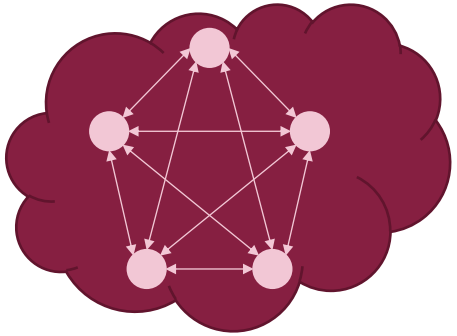
P2P



Leviathan



Intermediary



Blockchain

# Blockchain

An open network where everybody can establish trust themselves

- **Decentralized**: Trust distributed across multiple entities
- **Open**: Anyone can participate and verify the integrity and trustworthiness
- **Anonymity**: Everybody identity remains hidden
- **Enhanced confidentiality, integrity & privacy**
- **Eliminate centralized party**: Reduced operational cost

# Blockchain

Blockchain establish trust via software programs

- **Anonymity**
  - Everybody is equal and anonymous
  - Eliminate impacts of counterparty identify in justifying the trustworthiness
- **Decentralized** platform with <u>reward incentive</u> mechanisms
  - Encourage honest and trustworthy behavior of many participants
- **Smart contract** with predefined algorithms
  - Dispute resolution

# Blockchain

- In blockchain network, nothing to be trustworthy except its output
- All transactions validated via rigorous <u>mathematical proofs</u>
  - Anybody can verify proofs publicly

In Proof We Trust

# Why Blockchain?

- **Enhanced Security and Reliability**
  - via decentralization model and cryptography
  - Centralization more vulnerable to corruptions, errors, mistakes
- **Tamper-Proof**
  - Data alteration remains extremely difficult

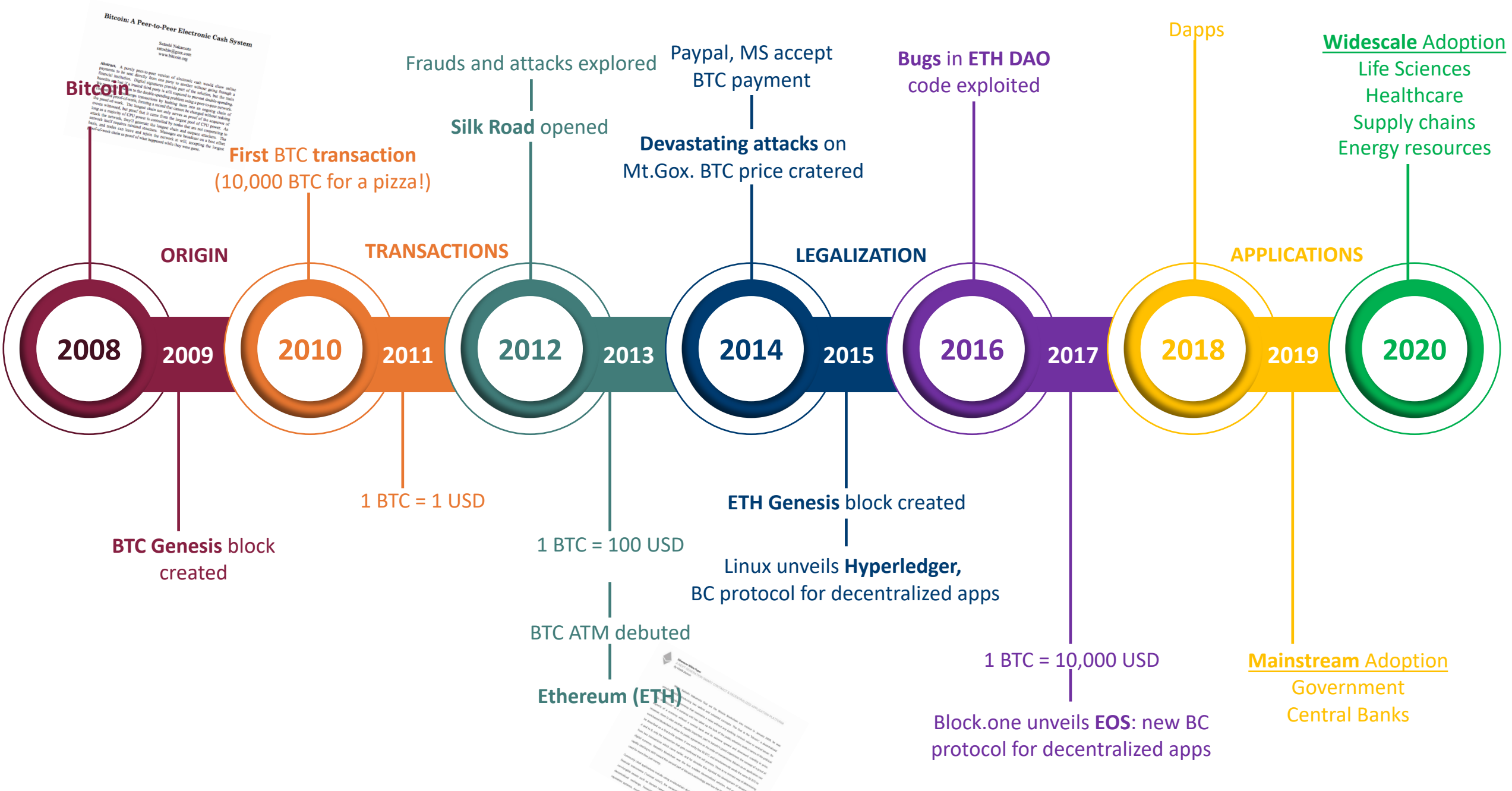
# Why Blockchain?

- **Digital Freedom**
  - Complete anonymity and transaction security and confidentiality
  - Transactions direct to recipients without routed to a trusted entity (e.g., bank)
  - Trusted entity always comes with risk and cost
- **Improved Transparency**
  - Everyone can verify and track transactions
  - Nobody can modify transactions on their own

# Why Blockchain?

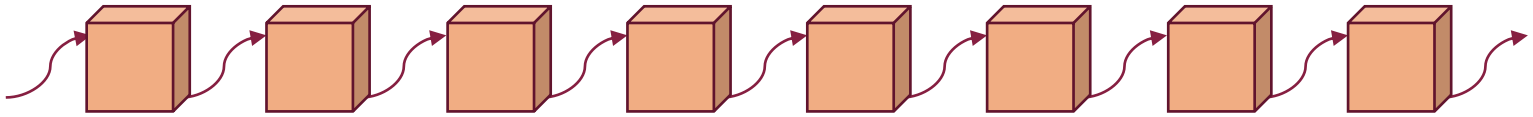
- **Better Efficiency and Reduced Cost**
  - Automated process
  - Minimal transaction cost
  - No need of maintaining central authority

# Blockchain Evolution



# What is Blockchain?

- First, Bitcoin is not Blockchain
  - Bitcoin is a digital currency that uses blockchain as the underlying data structure
- Blockchain is a data structure where data blocks are linked together
- Data blocks in the chain cannot be deleted or altered (**Immutability**)

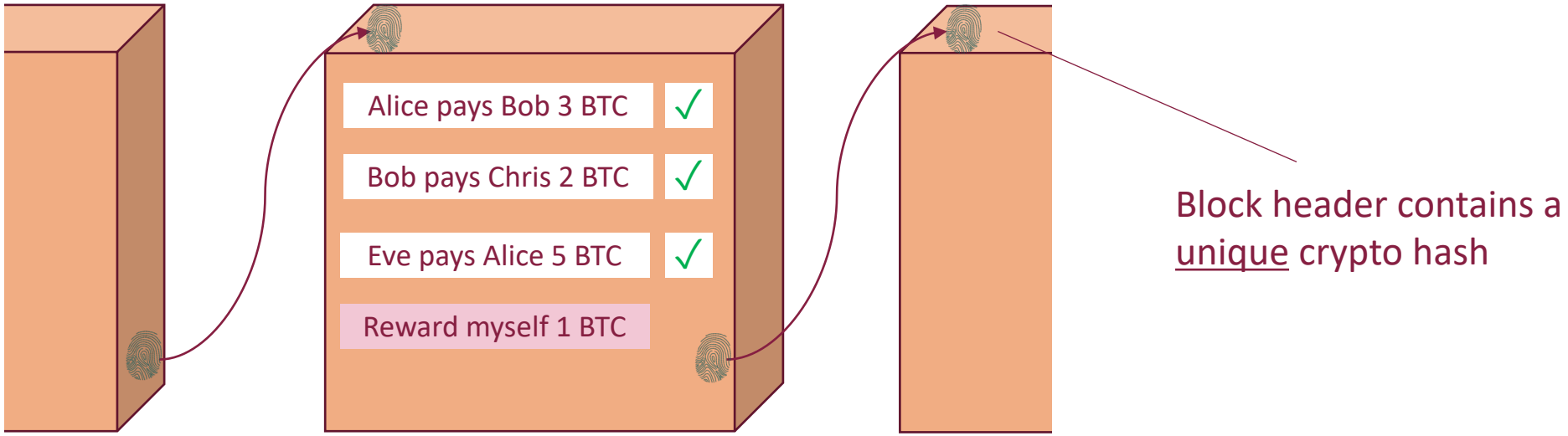


# What is Blockchain?

- Blockchain is a comprehensive system consisting of
  - Transactions
  - Immutable ledgers
  - Decentralized network
  - Data encryption/decryption
  - Consensus mechanisms
  - Smart contracts

# What is Blockchain?

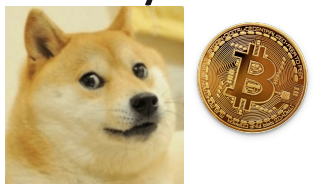
- Blockchain permits transactions to be gathered and recorded in the block
- Blocks are chained in chronological order via <u>cryptographic hash</u>



# How Blockchain Works?

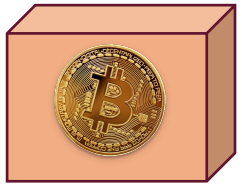
## In digital transaction…

D wants to send money to C



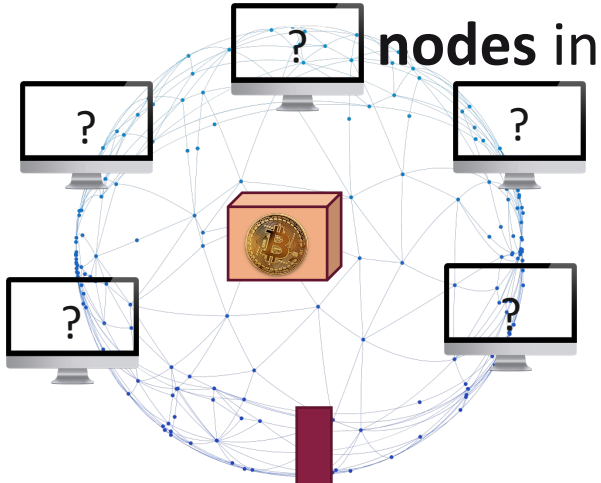
D

The **transaction** Tx is represented as a block

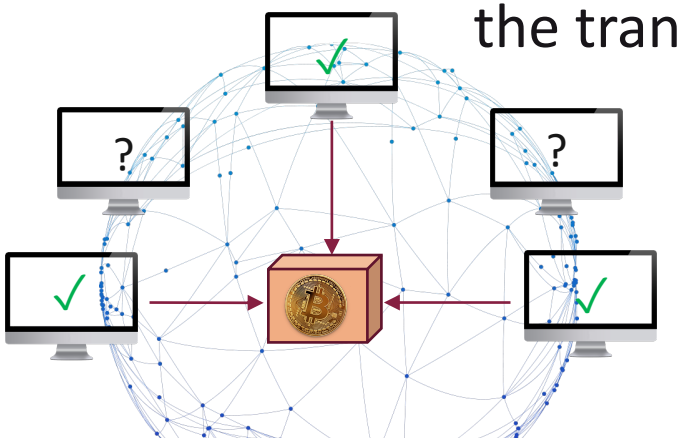


BC block

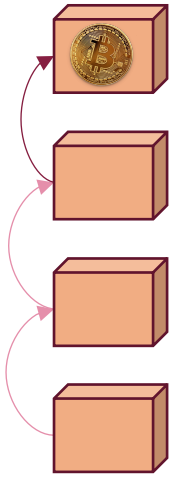
The block is broadcast to all the **distributed nodes** in the network




Sufficient nodes **verify and approve** the transaction




Tx is appended to Blockchain



BC

C receives the money

C

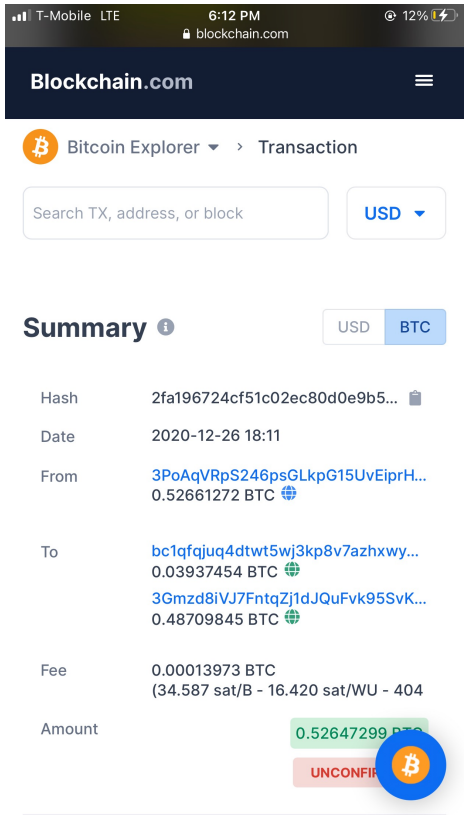
# Types of Blockchain

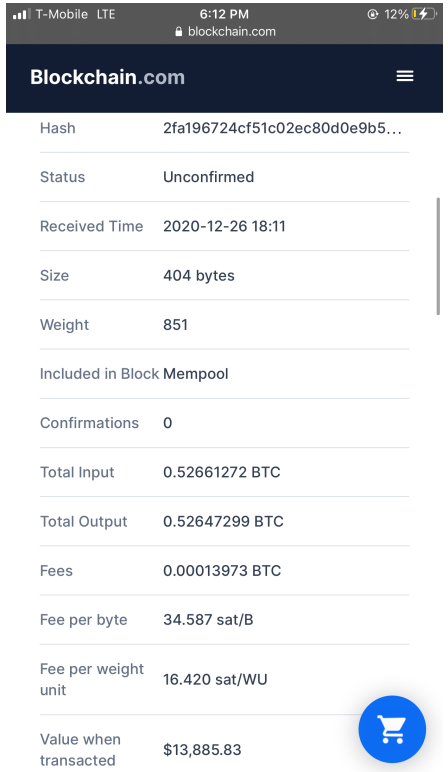
- **Public** and **private**
- Many mechanisms in public BC are not needed in private BC (handled by legal contracts)

| Characteristic | Public blockchain | Private blockchain |
|---|---|---|
| **Access** | Anyone can read/write | Only private group can read/write |
| **Authority** | Decentralized | Partially Decentralized |
| **Tx Speed** | Slow | Fast |
| **Consensus** | Permissionless | Permissioned |
| **Identity** | Anonymous | Known |
| **Efficiency** | Low | High |
| **Immutability** | Full | Partial |
| **Examples** | | |

# Blockchain Transactions

- Smallest element
- Record every decision and action taken
- Proof of history, provides provenance



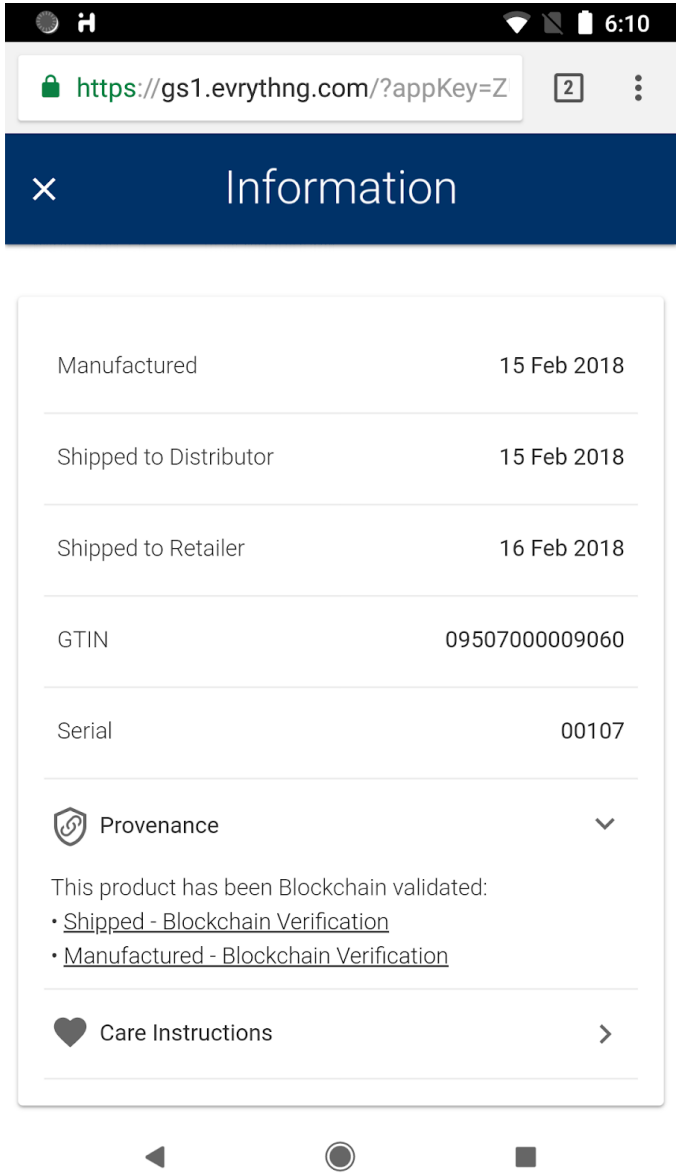
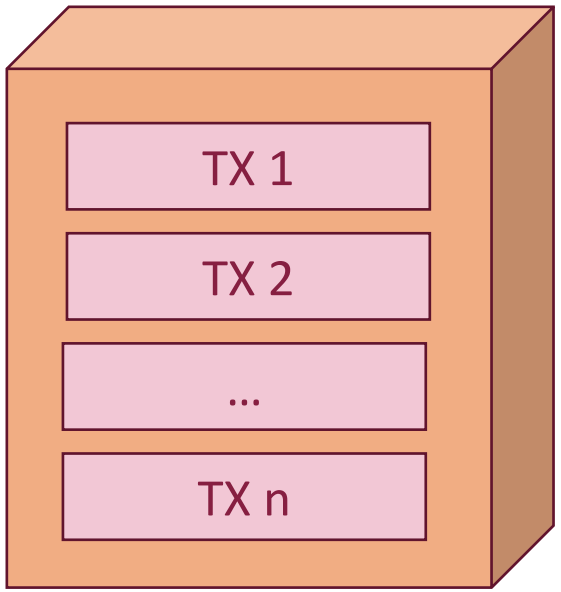




Image from https://evrythng.com/

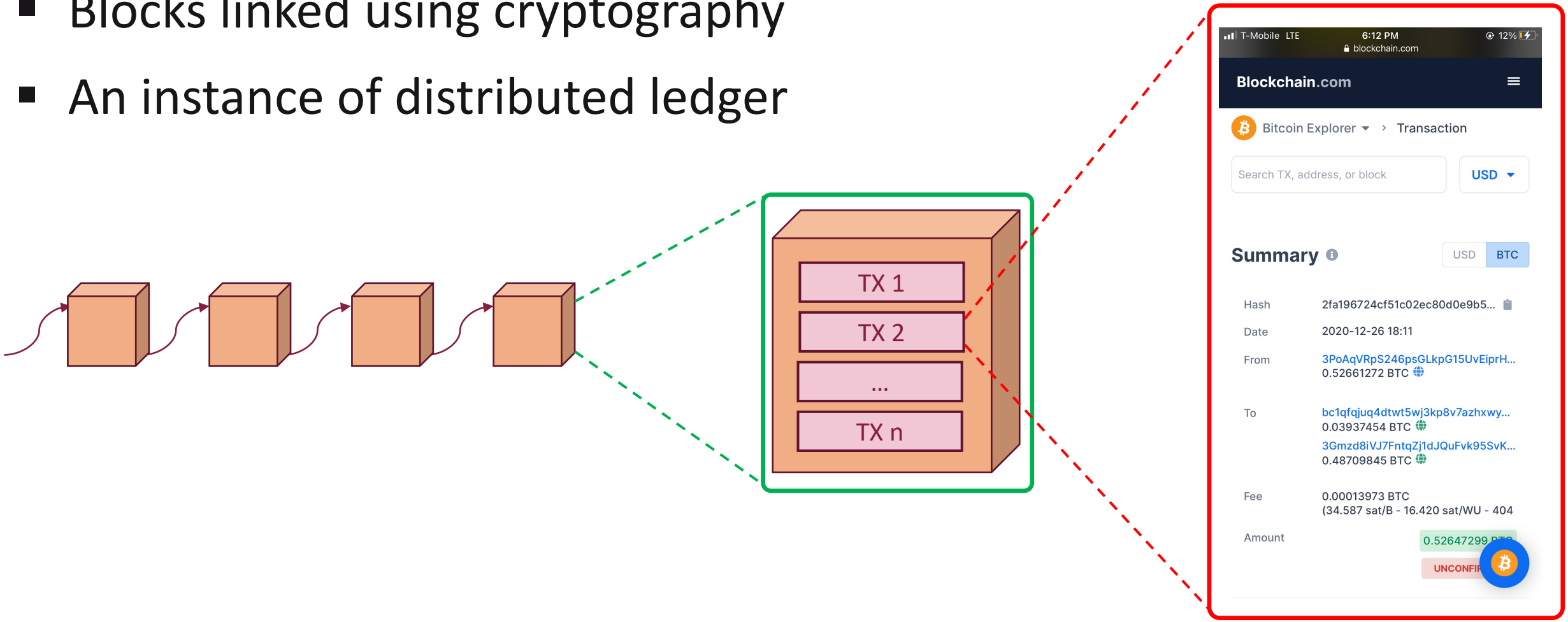
# Blockchain “Block”

- Contain multiple transactions
  - The transaction is immutable/indelible
- Write and Read-Only
- Once a block is chained, it is extremely difficult to change
  - Modification possible
  - Rework on all the subsequent blocks and consensus for each block



# Chain of Blocks

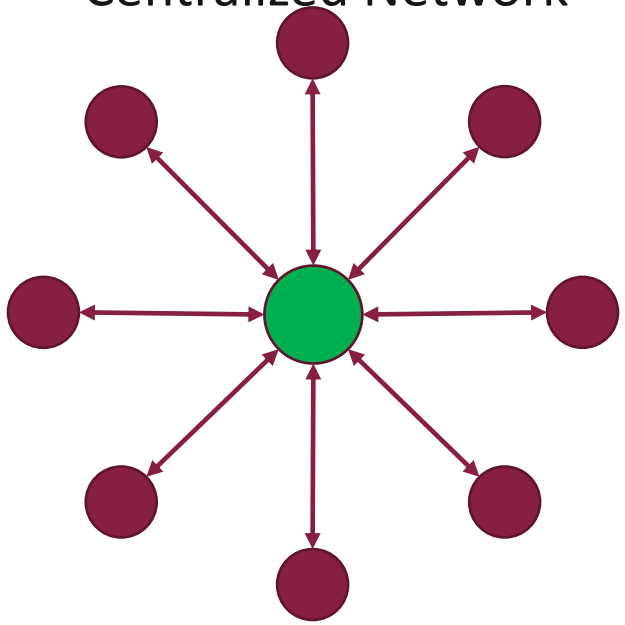
- Contain multiple blocks
- Blocks linked using cryptography
- An instance of distributed ledger



# Distributed Network

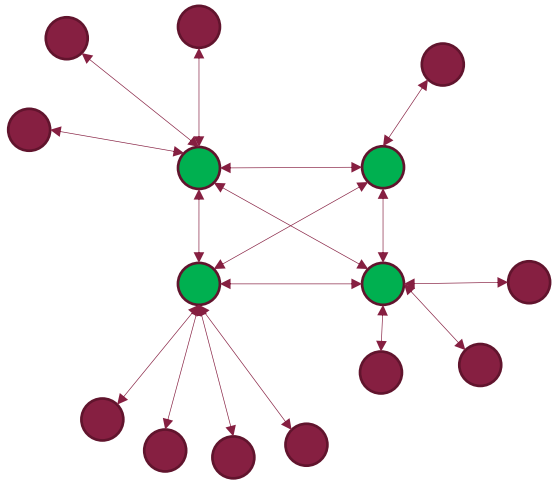
- Blockchain operates on a decentralized/distributed P2P network
- Each node stores a copy of the ledger
  - Distributed Ledger

Centralized Network



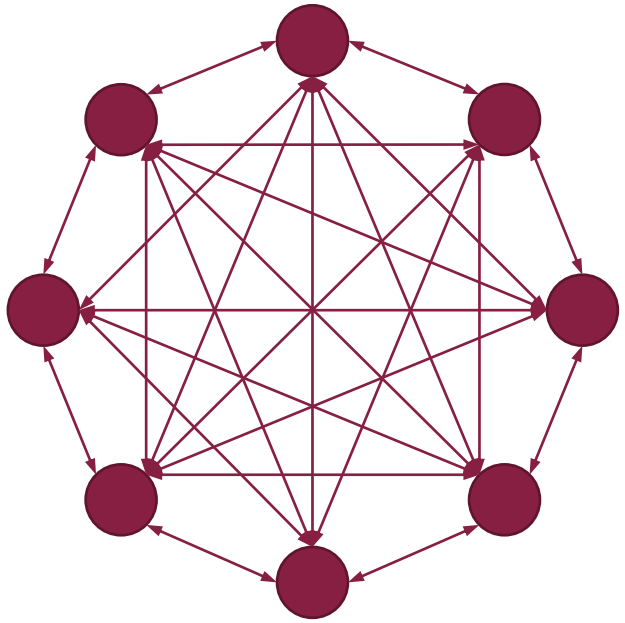
*nodes interact with a single central node*

Distributed Network



*nodes interact with some central nodes*

Decentralized Network

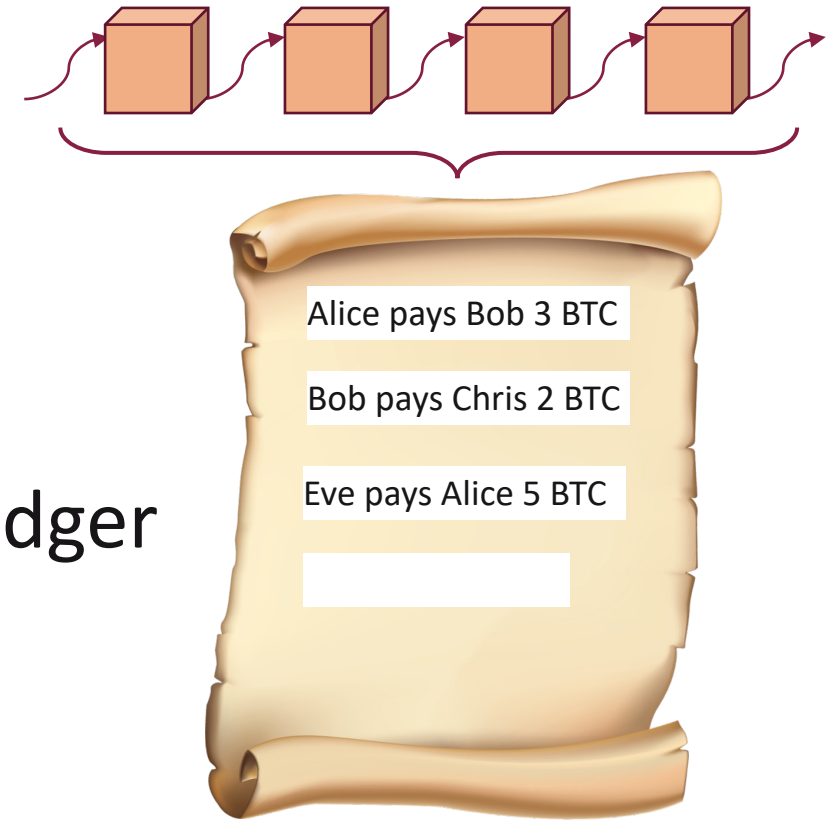


*Nodes interact with each other directly*

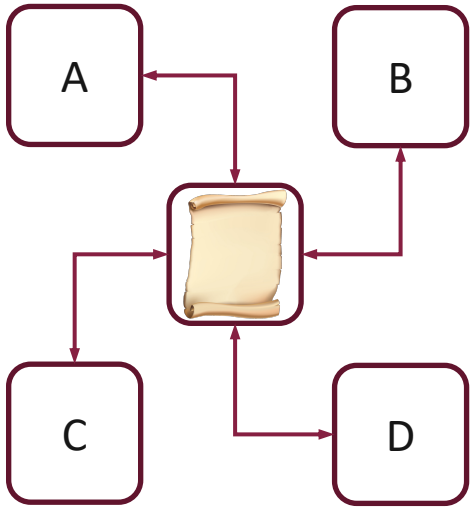
# Distributed Ledger

Blockchain is a <u>distributed ledger</u>

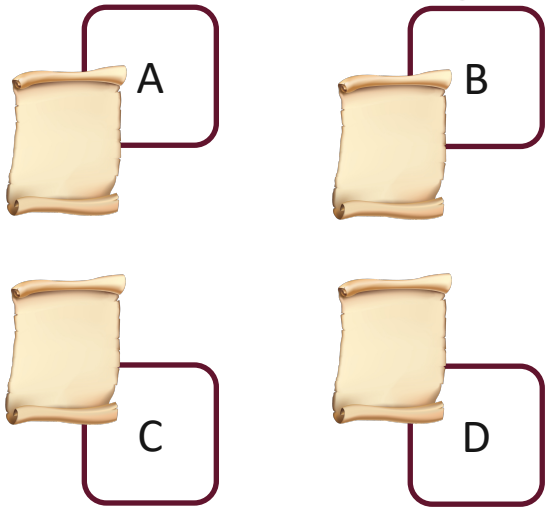
- **Centralized ledger:** stored by a central node
- **Distributed ledger**: stored in every node
  - All nodes agree on the true state of the ledger (via a consensus protocol)




Centralized ledger




Distributed ledger



# Distributed Ledger

- Keep track of all transactions performed in the network
- Can be encrypted for confidentiality
- Can be used without by individuals without a central authority
- **Immutable:** Ledger records are very difficult to be altered
  - Changing a record in the ledger requires a consensus from all participants
  - Rework on all subsequent records

# Distributed Consensus

- Ensure the blocks in blockchain are <u>valid</u> and <u>truthful</u>
- Prevent malicious adversaries from system compromise and chain-forking
- Many consensus protocols, each with different pros and cons
  - Proof of Work (PoW), Proof of Stake (PoS), Proof of Elapsed Time (PoET), Proof of Activity (PoA), Proof of Burn (PoB)
  - Paxos, BFT, Streamlet
- We will explore many of blockchain consensus protocols throughout this course

# Smart Contract

- A <u>program</u> running in <u>a secure environment</u> that controls the transfer of digital assets between parties under certain conditions
- Contract encoded into blockchain
- Enable broader blockchain applications beyond cryptocurrencies

```
/* Allow another contract to spend some tokens in your behalf */
function approve(address _spender, uint256 _value)
    returns (bool success) {
    allowance[msg.sender][_spender] = _value;
    return true;
}

/* Approve and then comunicate the approved contract in a single tx */
function approveAndCall(address _spender, uint256 _value, bytes _extraData)
    returns (bool success) {
    tokenRecipient spender = tokenRecipient(_spender);
    if (approve(_spender, _value)) {
        spender.receiveApproval(msg.sender, _value, this, _extraData);
        return true;
    }
}

/* A contract attempts to get the coins */
function transferFrom(address _from, address _to, uint256 _value) returns (bool success) {
    if (balanceOf[_from] < _value) throw;                 // Check if the sender has enough
    if (balanceOf[_to] + _value < balanceOf[_to]) throw;  // Check for overflows
    if (_value > allowance[_from][msg.sender]) throw;   // Check allowance
    balanceOf[_from] -= _value;                           // Subtract from the sender
    balanceOf[_to] += _value;                             // Add the same to the recipient
    allowance[_from][msg.sender] -= _value;
    Transfer(_from, _to, _value);
    return true;
}

/* This unnamed function is called whenever someone tries to send ether to it */
function () {
    throw;     // Prevents accidental sending of ether
```

Source: https://blockgeeks.com/guides/smart-contracts/

# Smart Contracts

- Smart contract is a computer program that
  - **Defines** rules
  - **Enforces** obligations and penalties
  - **Executes** actions required by clauses
  - **Autonomous** without ownership
  - **Secure**
- Written in a high-level programming language (e.g., Solidity)

| Blockchain Techniques | Smart Contracts? | Language |
|---|---|---|
| Bitcoin | ✗ | C++ |
| Ethereum | ✓ | **Solidity** |
| Hyperledger | ✓ | GoLang, C++, etc |

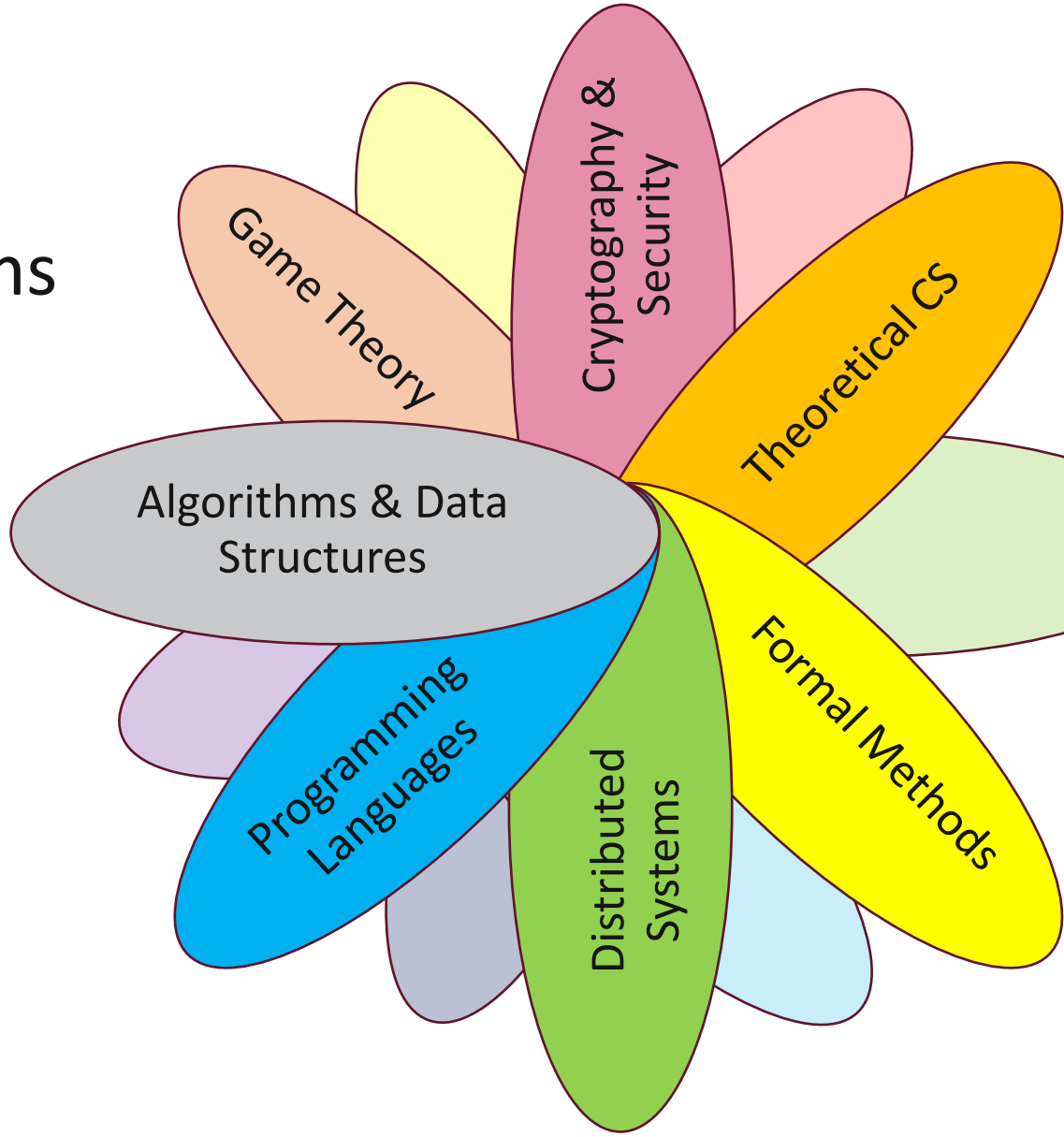
# Encryption

- Confidential transaction
- Prevent sensitive information to be leaked to malicious attacker
- Blocks can be partially or fully encrypted
  - Symmetric/asymmetric encryption
- Some (private) blockchains employ access control for visibility

# Challenges

- Energy consumption
- Resource-wasteful
- Immaturability
  - Scalability
  - Standardization
  - Awareness and Understanding
- Interaction with legacy infrastructure

# Summary

- Blockchain is interdisciplinary
- Cryptography and Distributed Systems are fundamental building blocks

| Operation | Crypto Techniques |
|---|---|
| Init & Broadcast Transactions | • Digital Signature<br>• Private/Public Keys |
| Transaction Validation | • Proof-of-Work |
| Chaining blocks | • Hash Function |